ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

6 question sets • 180 questions

CONTENTS

Set 1	**HSN-CN.A.1**	Write complex numbers in the form a + bi	3
Set 2	**HSN-CN.A.2**	Add and subtract complex numbers	40
Set 3	**HSN-CN.A.2**	Multiply complex numbers and simplify powers of i	77
Set 4	**HSN-CN.C.7**	Solve quadratic equations with complex solutions	114
Set 5	**HSN-CN.C.8**	Factor polynomials over the complex numbers	151
Set 6	**HSN-CN.C.9**	Use the Fundamental Theorem of Algebra	188

Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-49)}$ in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write $\sqrt{(-18)}$ in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write $(-4 + \sqrt{(-75)}) / 2$ in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write $(10 + \sqrt{(-64)}) / 4$ in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify $\sqrt{50}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify $\sqrt{72}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 2√3

B. 4√3

C. 3√2

D. 6√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify $\sqrt{28}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. -6

B. 18

C. $\sqrt{-36}$

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write $(8 + 6\sqrt{2}) / 2$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

Write $(-6 + 4\sqrt{5}) / 2$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

$(12 - 9\sqrt{3}) / 3$ equals

A. $4 - 9\sqrt{3}$

B. $4 - 3\sqrt{1}$

C. $12 - 3\sqrt{3}$

D. $4 - 3\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write $(-4 + 2\sqrt{3}) / (-2)$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and $(x - 3) + 2yi = 5 - 8i$. Find x and y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and $3a + (b + 1)i = -12 + 4i$. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and $(p + q) + (p - q)i = 10 + 2i$. What is p?

A. 5

B. 6

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for $-3 + 5i$.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point $(4, -2)$ in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. $-7i$

B. -7

C. $7 + i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify $(6 - 4i) - (-2 + 9i)$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify $(-1 + \sqrt{(-16)}) + (5 - \sqrt{(-9)})$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which $(4 - 7i) + w = -2 + 3i$.

$w =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify (3 + 5x) + (−7 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify $(6 - 4x) - (-2 + 9x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

$(2 - 3y) - (5 - y)$ equals

A. $7 - 4y$

B. $-3 + 2y$

C. $-3 - 4y$

D. $-3 - 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 - 4t.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

Simplify $(0.5 - x) - (1.5 + 2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-16)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

$\sqrt{(-25)}$ equals

A. -5

B. $25i$

C. $5i$

D. $-5i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is $(3 + ki) + (k - 5i)$ a real number? Give that real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
$(2k - 6) + 4i - (k + 1 - i)$ a pure imaginary number (real part 0)?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its opposite is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and $(a + 3i) - (2 + ai)$ is real. What is a?

A. 0

B. 3

C. 2

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

Explain why the sum of $a + bi$ and $a - bi$ is always a real number when a and b are real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

Plot 2 + i and 1 + 3i as points in the complex plane (real axis horizontal). What point represents their sum?

Sum point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral. What kind of quadrilateral is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points $5 + 2i$ and $1 - i$ in the complex plane?

A. 25

B. 5

C. $\sqrt{13}$

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

SET 3 • HSN-CN.A.2 • THE COMPLEX NUMBER SYSTEM

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(3 + 2i)(4 - 5i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(3 + 2x)(4 - 5x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

$(5 - 3x)(5 + 3x)$ equals

A. $25 - 9x^2$

B. $25 + 9x^2$

C. $10 - 9x^2$

D. $25 - 30x - 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply $x(4 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^9

B. i^8

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is $(a + 2i)(3 - i)$ a real number? Give the product.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 1$

B. $c = 0$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

Show that $(a + bi)(a - bi)$ is a real number for all real a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write $10 / (3 - i)$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 + 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. $-1 + 2i$

B. $1 - 2i$

C. $2 + i$

D. $1 + 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

Explain why multiplying by the conjugate of the denominator always gives a real denominator.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-CN.C.7 • THE COMPLEX NUMBER SYSTEM

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Write the quadratic formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

Write $\sqrt{(-48)}$ in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\sqrt{(-20)}$ equals

A. $2\sqrt{5}\,i$

B. $4\sqrt{5}\,i$

C. $-2\sqrt{5}$

D. $10i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write $(-2 + \sqrt{(-36)}) / 4$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 2)^2 = 9$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

Solve $x^2 = 50$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Write the equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Explain why $x^2 - (2 + i)x = 0$ has the solution $2 + i$ but not the solution $2 - i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

For which real values of c does $x^2 - 4x + c = 0$ have non-real solutions? Give them for c = 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when $m =$

A. 9

B. 3

C. 36

D. -9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For $c > 4$, write the solutions of $x^2 - 4x + c = 0$ in terms of c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 + 1)^2$

D. $(x^2 - 1)(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 + 2$

C. $x^2 - 2x$

D. $x^2 - 2x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q4**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 25 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

Solve $x^2 - 6x + 13 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. $4i$ and $-4i$

B. -4

C. $2i$ and $-2i$

D. 2 and -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

Solve $x^2 + 2x + 5 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - (-3/2 + (3\sqrt{3}/2)i)$

D. $x - 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and $-i$. Its third zero is

A. -5

B. $5i$

C. i

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Explain why a polynomial with real coefficients and degree 3 must have at least one real zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? Name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

List every complex zero of $p(x) = (x - 2)^3(x^2 + 9)$ with its multiplicity, and check that the multiplicities add to the degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Show that the Fundamental Theorem of Algebra holds for every quadratic $p(x) = ax^2 + bx + c$ with real a, b, c and $a \neq 0$: p has exactly two complex zeros counted with multiplicity.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$ and check that the count agrees with the Fundamental Theorem of Algebra.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Use the Fundamental Theorem of Algebra to explain how many complex zeros it has, and explain why at least one of them is real.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q5**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 2

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

Find the zeros of $(x - 2)(x + 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

Find the zeros of $(x - 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of $x(x + 4)$ are

A. 0 and 4

B. 0 and -4

C. -4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

Find the zeros of $(2x - 3)(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 9 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

Solve $x^2 + 2x + 2 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $2 \pm i$

D. $1 \pm 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

Solve $x^2 = -81$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

The graph of $y = x^3 - 4x^2 + 5x - 2$ touches the x-axis at $x = 1$ and crosses it at $x = 2$. List all complex zeros with multiplicity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 5

B. 2

C. 4

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9

Prove that every polynomial of odd degree with real coefficients has at least one real zero.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. What are all of its possible zero lists?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients may have no real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 7 only

B. 0, 2, 4, 6

C. 1, 3, 5, 7

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Explain why the Fundamental Theorem of Algebra says nothing about whether a polynomial's zeros can be found exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5